QBUS5001

Q&A Lecture notes - Week 8

Part 1 - Introduction to hypothesis testing

In Week 1, we learned that inferential statistics is all about using sample **stasitics** to estimate population **parameters**.

List of sample statistics and population parameters:

	Population parameter	Sample statistic
Mean	μ	$\bar{X}$
Proportion	p	$\hat{p}$

Last week, we talked about interval estimation. This week, we'll look at another technique in estimation – hypothesis testing.

The logic behind hypothesis testing (and terminologies)

1. Setting Up Hypotheses

When conducting a hypothesis test, we begin by making two opposing hypotheses about the population parameters. These are called the **Null hypothesis** and the **alternative hypothesis**. The value stated in the null hypothesis is referred to as the **hypothesised population parameter value**.

For example, if your null hypothesis states that H_0: $\mu = 10$, we interpret it as:

"The hypothesised population mean is 10."

The null hypothesis represents the current belief or assumption about the population parameter, and the alternative hypothesis challenges this assumption.

2. Testing the Hypotheses Using Evidence from a Sample

Once the hypotheses are set, we collect data from a sample and use the **observed sample statistic** as evidence. This evidence helps us decide whether to reject or not reject the **Null hypothesis**.

- If the observed sample statistic is **not very far** from the hypothesised population parameter, we **do not reject the null hypothesis**
- If the observed sample statistic is **significantly far** from the hypothesised population parameter, we **reject the null hypothesis**.

Example:

$H_0: \mu = 5$

$\bar{X} = 5.01$

$\bar{X} = 3.2$

Remember: We never say "we accept the null hypothesis." This is because, even if we do not have enough evidence to reject the null hypothesis, it does not mean that the null hypothesis is correct. We can only state that there is insufficient evidence to reject it.

1.96

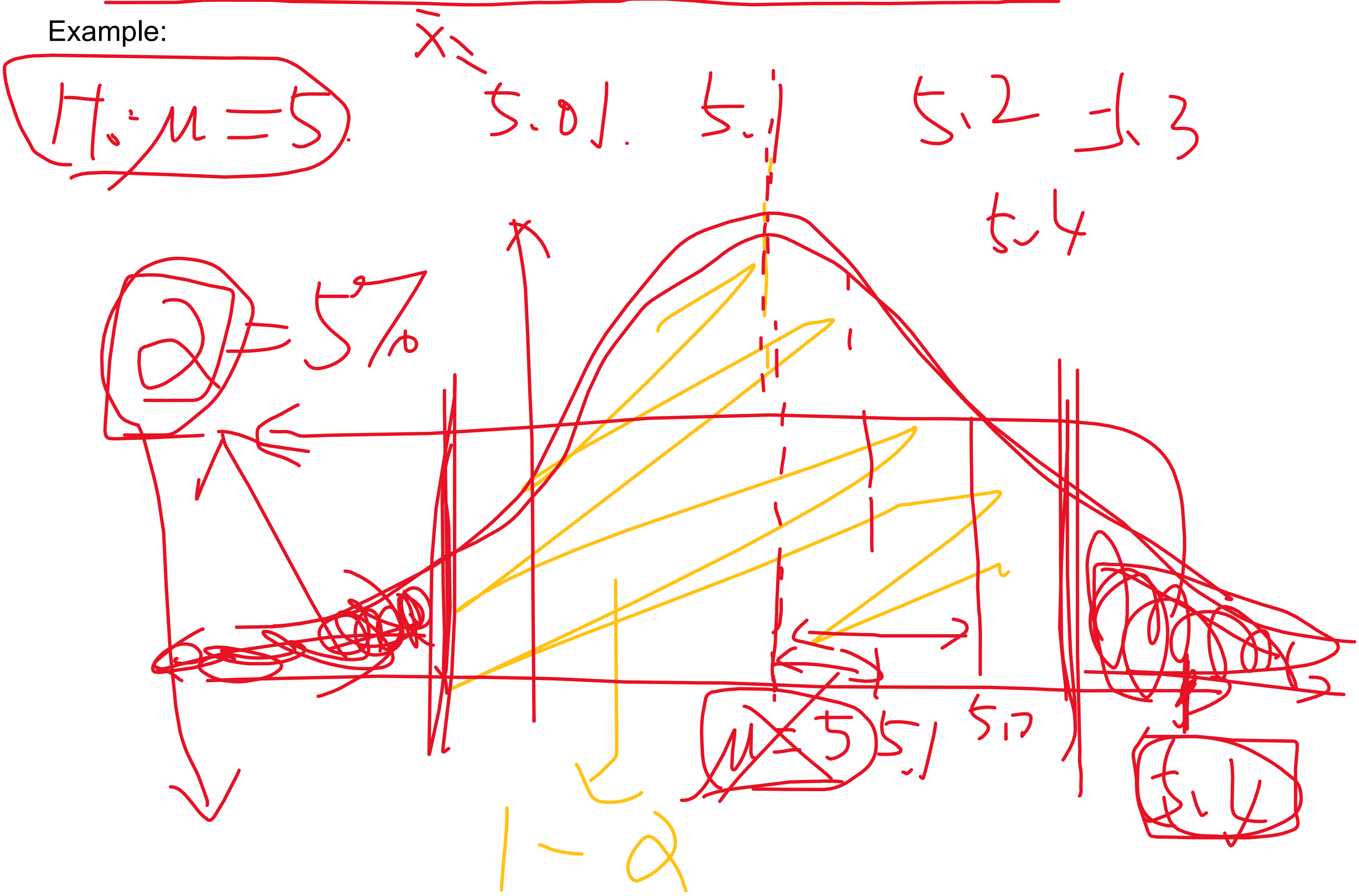
3. Deciding How Far is "Far Enough"

Now, the key question is: *How far away does the sample statistic need to be from the hypothesised population parameter for us to reject the null hypothesis?*

To determine this, we rely on **probabilities** and the **sampling distribution** of the statistic. These tools help us quantify the likelihood of observing such a result if the null hypothesis were true.

In the next section, we will walk through the step-by-step process of hypothesis testing, using these concepts to guide you in making accurate conclusions based on your data.

Example:



MCQ

4. Level of Significance – Controlling the Probability of Making a Wrong Decision

We understand that hypothesis testing involves the risk of making a wrong decision, but we control this risk by setting a **level of significance**.

- **Significance Level and Type I Error**:
  Last week, we discussed that the **confidence level** is denoted by **1 - α**. Here, **α** is called the **significance level**, representing the probability of erroneously rejecting the null hypothesis when it is actually true (this mistake is called a **Type I error**). Importantly, **α** is chosen by us, often set at 0.05 or 5%.
- **Type II Error**:
  There's another probability, denoted **β**, which is the likelihood of failing to reject the null hypothesis when it is actually false (this is called a **Type II error**).
- **Power of the Test**:
  The **power of the test** is denoted by **1 - β**, which represents the probability of correctly rejecting the null hypothesis when it is false. A high power means that the test is more likely to detect an actual effect or difference.

Now let's summarise everything in this table:

	H_0 is correct	H_0 is not correct
Reject H_0	Type I error P(*)=α Significance level	Right $P(*) = 1 - \beta$ Power of the test
Do not reject H_0	Right P(*)=1-α	Type II error P(*)=β

Relationships between α and β

- With the **same sample size**, increasing **α** (significance level) decreases **β** (Type II error), and vice versa.
- **Increasing the sample size** reduces both **α** and **β**, enhancing the accuracy of the hypothesis test.

Part 2 - Hypothesis testing for the mean

Let's get started by reviewing what we've done in week 5.

Sampling distribution of sample mean:

If population follows normal distribution, sampling distribution of sample mean is ___________(**always**/sometimes/approximately) normally distributed.

If the population is non-normal (any distribution), the sampling distribution of sample mean is approximately normally distributed if the sample size is large (n>30). This is called _________________(**central limit theorem**/central unlimited theorem/side limit theorem)

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

This is equivalent to the standardised sample mean

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0,1)$$

Particularly, when the population standard deviation (variance) is unknown, the sampling distribution of the standardised sample mean follows t-distribution.

$\bar{X} = 5.1$

$H_0: \mu = 5$

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{n-1}$$

In this week, the standardised sample statistic has a new name – **test statistic**!

Why?

Suppose we take a sample and observe the sample mean $\bar{x}$ and sample sd s

60 → 1 min

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

$\frac{360s - 240s}{60}$

$$\bar{x} \rightarrow observed\ sample\ mean\ from\ the\ sample$$

$$\bar{x} - \mu \rightarrow the\ difference\ between\ observed\ sample\ mean\ and\ the\ hypothesised\ population\ mean$$

Important understanding:

A test statistic describes how ~~many standard~~ errors between the observed sample statistic and the hypothesised population parameter

$\bar{x}$

μ

$$\frac{\bar{X}-\mu}{SE}$$

Now, let's walk through a real-life example, step by step, to understand how a hypothesis test is performed. Along the way, we will also learn some important terminologies related to hypothesis testing.

Example 1: Testing the Weight of Sugar at Chatime with known population standard deviation

Gary owns Chatime at Darling Square, and he is very particular about the quality of the bubble tea his shop produces. His staff claim that the **average weight of sugar** used for their signature product, *Chai Latte*, is $\boxed{\textbf{5 grams}}$. This weight ensures a balanced sweetness while keeping the cost manageable.

μ n σ

Gary decides to test this claim. It is known that the population **standard deviation** is **0.6 grams**. He takes a **random sample** of 36 Chai Lattes and measures the average weight of sugar used. The sample shows an average weight of **4.8 grams**. → $\bar{x}$

Using a **5% level of significance**, we will now test whether the average weight of sugar used is truly **5 grams**.

→ α

μ

Step 1: setting up null and alternative hypothesis

$$H_0: \mu = 5$$

$$H_1: \mu \neq 5$$

~~$H_0: \bar{X} = 5$~~

~~$H_1: \bar{X} \neq 5$~~

~~$H_0 = 5$~~

~~$H_1 \neq 5$~~

Attention!

Tony's secret weapon to set up null and alternative hypothesis

1. read the question "test xxxxxxxxxx xxxx...." carefully

2. The claim usually splited the sample space into two parts. Write the two statements down regarding the population parameter.

$$\mu = 5$$

$$\mu \neq 5$$

3. Put whichever of the two statements that has a "=" into the null hypothesis.

$$H_0: \mu = 5$$

$$H_1: \mu \neq 5$$

Example 1.2: test if the average weight of sugar is less than 5.

$$H_0: \mu \geq 5 \; or \; \mu = 5$$

$$H_1: \mu < 5$$

Example 1.3: test if the average weight of sugar is no more than 5.

$$H_0: \mu \leq 5$$

$$H_1: \mu > 5$$

Step 2 – specifying the sample size and level of significance

$$n = 36$$

$$\alpha = 0.05$$

Step 3 – Specifying the test statistic and its sampling distribution. Make assumptions when necessary (Write down the formula you are going to use, with underlying assumptions)

Under H_0, as the sample size = 36 > 30, CLT applies, the test statistic is given by

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0,1) \; approximately$$

What if the sample size is not large?

Under H_0, as the sample size = 25 < 30, CLT does not apply. We need assume the population sugar level follows a normal distribution. So the test statistic is given by

$$\boxed{Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}} \sim N(0,1)$$

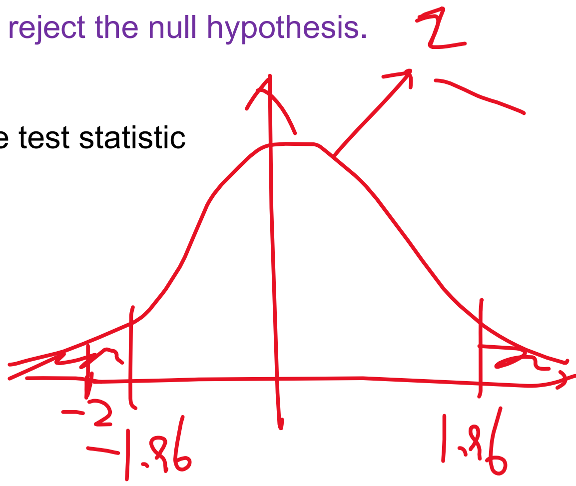
Step 4 – State the decision rules (rejection region approach)

Purpose of this step: given 5% level of significance, determine how many standard errors away is considered far away enough to reject the H_0

At 5% level of significance, H_0 is rejected if the observed value of the test statistic

$$|z_{obs}| > z_{0.025} = 1.96$$

Interpretation: 1.96 standard errors away between the observed sample mean and the hypothesised population mean is considered far away enough to reject the null hypothesis.



Step 5 – collect a sample and calculate the observed value of the test statistic

Now, the observed value of the test statistic is

$$z_{obs} = \frac{4.8 - 5}{\frac{0.6}{\sqrt{36}}} = -2 < -1.96$$

Step 6 – Draw a conclusion

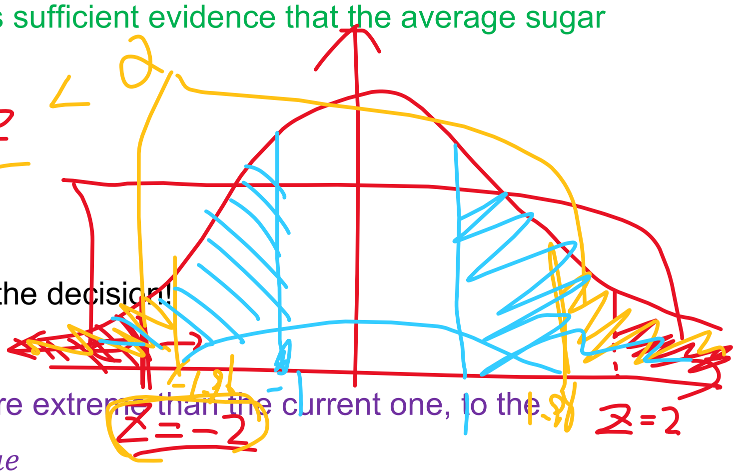
Golden template in drawing a conclusion

H_0 [is/ is not] rejected at α level of significance, and there is [sufficient/insufficient] evidence that [whatever is in H_1]

In this question

H_0 is rejected at 5% level of significance, and there is sufficient evidence that the average sugar level is not 5 grams.

p-value $< \alpha$



Alternatively, we can use p-value approach to make the decision!

What is a p-value?

The probability of observing a test statistic that is more extreme than the current one, to the direction of the alternative hypothesis, given H_0 *is true*

Step 4 – decision rules (p-value approach)

At 5% level of significance, H_0 is rejected if

$$p - value = P(Z < -2) + P(Z > 2) < \alpha$$

Step 5 – calculate the p-value

$$p - value = 2 \times Norm.s.dist(-2, TRUE)$$

Step 6 – Draw a conclusion.

H_0 is rejected at 5% level of significance, and there is sufficient evidence that the average sugar level is not 5 grams.

Example 2: Testing the weight of sugar with unknown population standard deviation

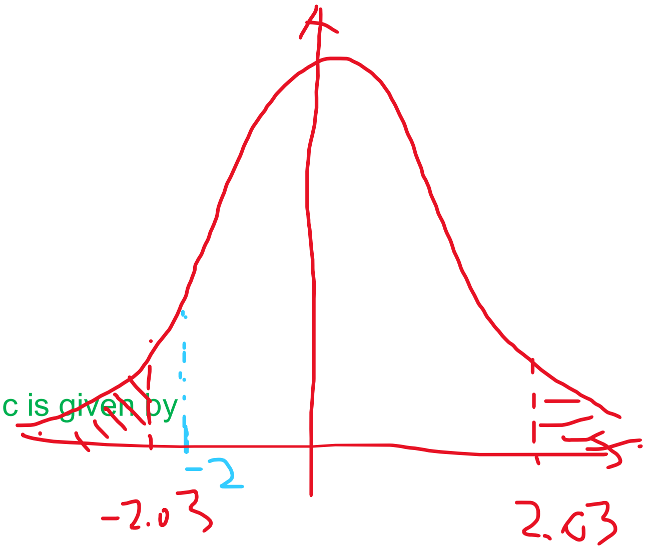
Gary owns Chatime at Darling Square, and he is very particular about the quality of the bubble tea his shop produces. His staff claim that the **average weight of sugar** used for their signature product, *Chai Latte*, is **5 grams**. This weight ensures a balanced sweetness while keeping the cost manageable.

Gary decides to test this claim. He takes a **random sample** of 36 Chai Lattes and measures the average weight of sugar used. The sample shows an average weight of **4.8 grams**, and the sample sample **standard deviation** is **0.6 grams**.

Using a **5% level of significance**, we will now test whether the average weight of sugar used is truly **5 grams**.

(Use $t_{0.025,35} = 2.03 \ or\ find\ it\ by = t.inv.2t(0.05,35)$)

Now let's see the difference!



Test

$$H_0: \mu = 5$$

$$vs\ H_1: \mu \neq 5$$

Under H_0, as the sample size $n = 36 > 30$, the test statistic is given by

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{35}$$

At 5% level of significance, the Null hypothesis is rejected if the observed value of the test statistic

$$|t_{obs}| > t_{0.025,35}$$

Now, the observed value of the test statistic is

$$t_{obs} = \frac{\bar{x} - \mu}{s/\sqrt{n}} = \frac{4.8 - 5}{0.6/\sqrt{36}} = -2 > -t_{0.025,35} = -2.03$$

Therefore, we do not reject the H_0 at 5% level of significance, and we do not have enough evidence that the population average weight of the sugar content is not 5 grams.

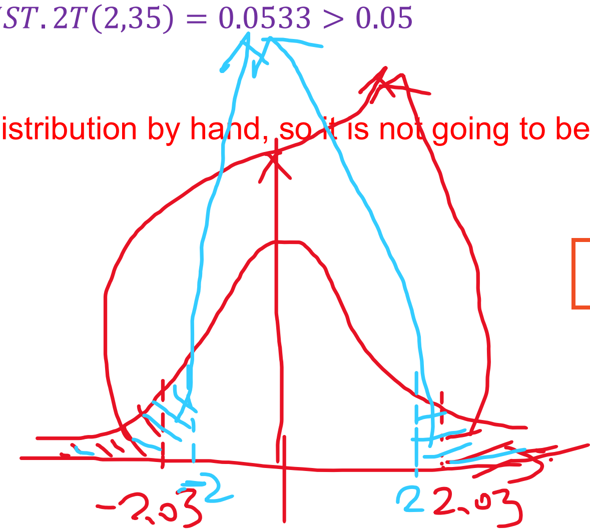
***p-value approach**

The p-value of this test is given by

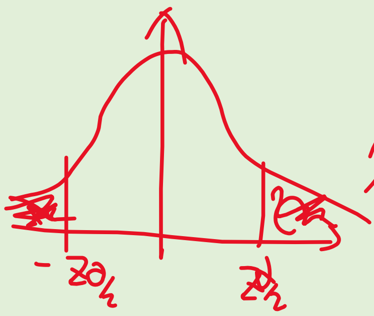
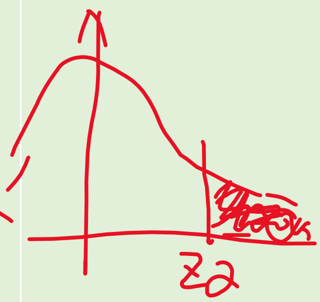
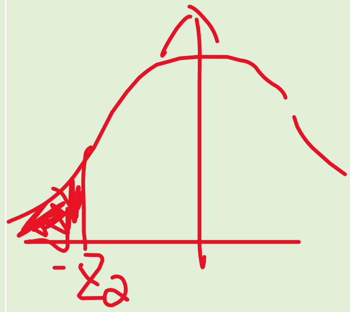
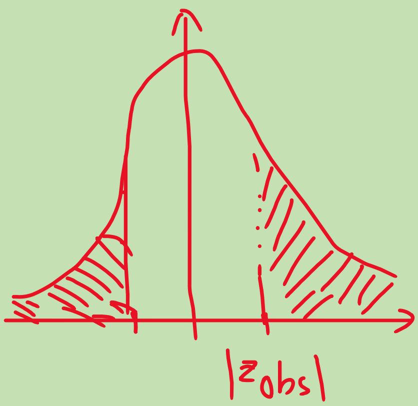
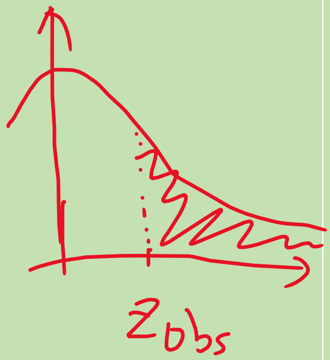
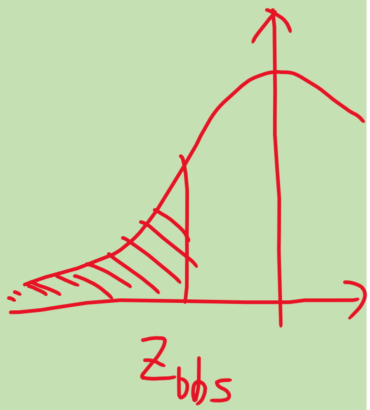
$$p - value = P(T > 2) + P(T < -2) = T.DIST.2T(2,35) = 0.0533 > 0.05$$

Same conclusion is drawn.

Good news: it is impossible to calculate p-value for a t distribution by hand, so it is not going to be tested.



Summary of decision rules on one-tailed and two-tailed test

	Two-tailed test	**Upper-tailed test**	**Lower-tailed test**
Indicator (alternative hypothesis)	$H_0::\mu = value$ $H_1:\mu \neq value$ Test if population parameter is equal to {value}	$H_0::\mu = value$ $H_1:\mu > value$ Test if population parameter is greater than {value}; is no great than {value}	$H_0::\mu = value$ $H_1:\mu < value$ Test if population parameter is less than {value}; is no less than {value}
Decision rule Critical value approach	Reject H0 if $\lvert z_{obs}\rvert > z_{\frac{\alpha}{2}}$  	Reject H0 if $z_{obs} > z_{\alpha}$  	Reject H0 if $z_{obs} < -z_{\alpha}$ 
p-value	$2 \times P(Z > \lvert z_{obs}\rvert)$  	$P(Z > z_{obs})$  	$P(Z < z_{obs})$ 
Decision rule on p-value	H0 is reject if p-value<α		
Note: all the zs and Zs in the table can be replaced by t or T; μ can be replaced by p			

Example 3: (For self study) One-sided test - Lower tailed test

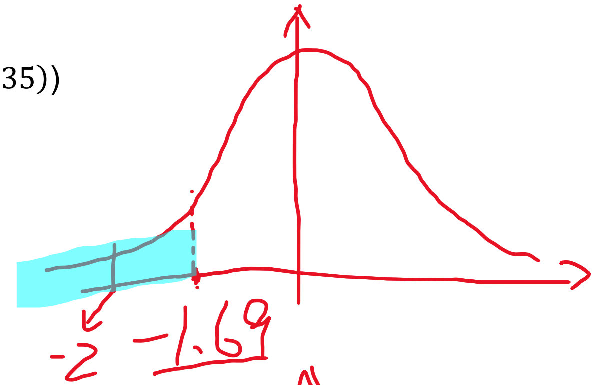
Gary owns Chatime at Darling Square, and he is very particular about the quality of the bubble tea his shop produces. His staff claim that the average weight of sugar used for their signature product, Chai Latte, is 5 grams. This weight ensures a balanced sweetness while keeping the cost manageable.

Gary suspects that his staff may be using **less than 5 grams** of sugar on average and decides to test this claim. He takes a random sample of 36 Chai Lattes and measures the average weight of sugar used. The sample shows an average weight of 4.8 grams, and the sample standard deviation is 0.6 grams.

Using a 5% level of significance, we will now test whether the average weight of sugar used is **less than 5 grams**.

(Use $-t_{0.05,35} = -1.69, or\ you\ can\ find\ it\ by = t.inv(0.05{,}35)$)

Now let's see the difference!



Test

$$H_0: \mu = 5$$

$$vs\ H_1: \mu < 5$$

Under H_0, as the sample size $n = 36 > 30$, the test statistic is given by

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{35}$$

At 5% level of significance, the Null hypothesis is rejected if the observed value of the test statistic

$$t_{obs} < -t_{0.05,35}$$

Now, the observed value of the test statistic is

$$t_{obs} = \frac{\bar{x} - \mu}{s/\sqrt{n}} = \frac{4.8 - 5}{0.6/\sqrt{36}} = -2 < -t_{0.05,35} = -1.69$$

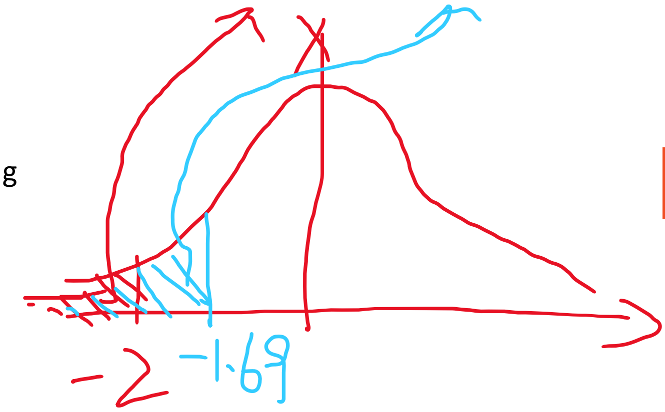
Therefore, we reject the H_0 at 5% level of significance, and we have enough evidence that the population average weight of the sugar content is less than 5 grams.

***p-value approach**

The p-value of this test is given by

$$p - value = P(T < -2) = T.DIST(-2{,}35, TRUE) = 0.0267 < 0.05$$

Same conclusion is drawn.



Example 4: (For self study) One-sided test - Upper tailed test

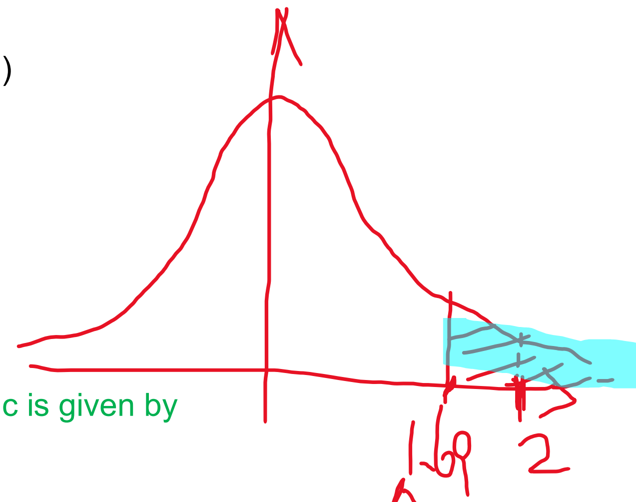
Gary owns Chatime at Darling Square, and he is very particular about the quality of the bubble tea his shop produces. His staff claim that the average weight of sugar used for their signature product, Chai Latte, is 5 grams. This weight ensures a balanced sweetness while keeping the cost manageable.

Gary suspects that his staff may be using **more than 5 grams** of sugar on average and decides to test this claim. He takes a random sample of 36 Chai Lattes and measures the average weight of sugar used. The sample shows an average weight of 5.2 grams, and the sample standard deviation is 0.6 grams.

Using a 5% level of significance, we will now test whether the average weight of sugar used is **more than 5 grams**.

(Use $t_{0.05,35} = 1.69, which\ is\ found\ by = t.inv.rt(0.05,35)$)

Now let's see the difference!



Test

$$H_0: \mu = 5$$

$$vs\ H_1: \mu > 5$$

Under H_0, as the sample size $n = 36 > 30$, the test statistic is given by

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{35}$$

At 5% level of significance, the Null hypothesis is rejected if the observed value of the test statistic

$$t_{obs} > t_{0.05,35}$$

Now, the observed value of the test statistic is

$$t_{obs} = \frac{\bar{x} - \mu}{s/\sqrt{n}} = \frac{5.2 - 5}{0.6/\sqrt{36}} = 2 > t_{0.05,35} = 1.69$$

Therefore, we reject the H_0 at 5% level of significance, and we have enough evidence that the population average weight of the sugar content is greater than 5 grams.

***p-value approach**

The p-value of this test is given by

$$p - value = P(T > 2) = T.DIST.RT(2,35) = 0.0267 < 0.05$$

Same conclusion is drawn.

Part 3 - Hypothesis test for the proportion

Again let's start by reviewing what we've got in week 5

Sampling distribution of sample proportion

If $np > 5$_ and $n(1-p) >$ _5, the sampling distribution of sample proportion can be approximated by a normal distribution.

$$\hat{p} \sim N\left(p, \frac{p(1-p)}{n}\right)$$

Or the standardised $\hat{p}$ is also the **test statistic**

$$Z = \frac{\hat{p} - p}{\sqrt{\frac{p(1-p)}{n}}} \sim N(0,1)$$

Example 5

Team Illusion is renowned for taking on a variety of challenges, and its captain, URE, is known for successfully completing the majority of them. It is claimed that URE succeeds in **more than 80%** of the challenges.

To test this claim, a random sample of 100 challenges was taken, and it was found that URE successfully completed 83 of them.

Using a **5% level of significance**, is there enough evidence to support the claim that URE's success rate is **higher than 80%**?

(Use $z_{0.05} = 1.645\ or\ find\ it\ by = norm.s.inv(0.95)$)

Test

$$H_0: p = 0.8$$

$$H_1: p > 0.8$$

Under H_0, since $np = 0.8 \times 100 = 80 > 5$, $n(1-p) = 20 > 5$, the test statistic is given by

$$Z = \frac{\hat{p} - p}{\sqrt{\frac{p(1-p)}{n}}} \sim N(0,1)$$

At 5% level of significance, the Null hypothesis is rejected if the observed value of the test statistic

$$z_{obs} > z_{0.05} = 1.645$$

Now, the observed value of the test statistic is

$$z_{obs} = \frac{0.83 - 0.8}{\sqrt{\frac{0.8 \times 0.2}{100}}} = 0.75 < 1.645$$

Therefore, we do not reject the H_0 at 5% level of significance, and we do not have enough evidence that the population proportion of success rate is more than 80%.

Summary of everything in week 6 and 8

Estimating population mean μ		
Case	**σ is known**	**σ is unknown**
Point estimator	$\bar{X}$	
Expected value of the point estimator	μ	
Standard error of the point estimator	$\frac{\sigma}{\sqrt{n}}$	$\frac{S}{\sqrt{n}}$
Sampling Distribution of the standardised point estimator (aka. Test statistic)	$\frac{\bar{X}-\mu}{\frac{\sigma}{\sqrt{n}}} \sim N(0,1)$	$\frac{\bar{X}-\mu}{\frac{S}{\sqrt{n}}} \sim t_{n-1}$
Assumptions or decisions to have the above sampling distribution	if n>30, central limit theorem applies, no assumptions needed. If n<30, central limit does not apply, so we need to assume that the population follows a normal distribution.	
Confidence interval estimator	$\bar{X} \pm z_{\alpha/2} \times \frac{\sigma}{\sqrt{n}}$	$\bar{X} \pm t_{\frac{\alpha}{2},n-1} \times \frac{S}{\sqrt{n}}$

Sample size determination	$$n = \left(\frac{Z_{\frac{\alpha}{2}}\sigma}{e}\right)^2$$ Where can we get σ? 1. σ can be estimated by $\frac{range}{4}$ 2. We may conduct a pilot study and estimate σ by S.

Estimating population proportion p	
Case	Only one case
Point estimator	$\hat{p}$
Mean of the point estimator	p
Standard error of the point estimator	$\sqrt{\frac{p(1-p)}{n}}$
Sampling Distribution of the standardised point estimator (aka. Test statistic)	$Z = \frac{\hat{p}-p}{\sqrt{\frac{p(1-p)}{n}}} \sim N(0,1)\ approximately$
Assumptions to have the above sampling distribution	$np \geq 5$ $n(1-p) \geq 5$
Confidence interval estimator	$\hat{p} \pm z_{\alpha/2} \times \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$
Sample size determination	$n = \frac{z_{\alpha/2}^2 \times p(1-p)}{e^2}$ Where do we get p? 1. To be conservative, we may assign $p = 0.5$ 2. We may conduct a pilot study and estimate p using $\hat{p}$